Grade 12 Core Mathematics

Teachers' Book

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INNODEMS

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Preface

Introduction. This is a free interactive digital textbook following the Grade 12 Kenyan competency based maths curriculum and has been co-developed by INNODEMS, IDEMS and the Kenyan Maths Society. It comes with teaching resources such as lesson plans and step-by-step guides to support teachers teaching this curriculum for the first time. You can find these in the respective sections of the book.

Interactive elements of the online book include e.g. automatically evaluated checkpoint questions (see further below) and animations (TODO) that can be projected in class or used by learners at home to explore on their own.

This is the print version. You can find the interactive online version here: Online version.

Support. We also offer a free online course on teaching using competency-based education methodologies and contributing to these learning resources: CBE Math Delivery I Course

Access the course directly using the link above, or follow this guide to install the Moodle App to more easily follow the course on your phone.

We also offer free online trainings and opportunities to exchange you experience with other teachers. The strand-specific courses are linked in the sections of this book. You can find more information in the resources section of the project website.

For Learners. Click here for an (online) learner version of the book. Share this with learners (or their parents) who have access to digital devices. It features interactive **Checkpoint** questions that learners can use to practice at their own pace and get instant feedback on their solutions. These questions are randomized, so that learners can attempt a question multiple times and get a different random variant each time.

Checkpoints: An example. Below you can find an example of a checkpoint question. While in print these look like regular exercises, in the online version these are interactive. Learners can enter their answers in the input fields and get immediate feedback. They can also attempt new random variants of the questions to practice a given concept over and over.

Checkpoint 0.0.1 Exercise

Solve the quadratic equation $x^2 - x - 2 = 0$. If there is a repeated root, enter this twice.

$x =$ _____ or $x =$ _____

Answer 1. 2

Answer 2. -1

Solution. We are asked to factorise the quadratic equation $x^2 - x - 2 = 0$.

First factorise the left-hand side:

$$x^2 - x - 2 = (x - 2)(x + 1).$$

Equate each factor to zero to obtain the two solutions:

Solution 1: $x = 2$

Solution 2: $x = -1$.

Depending on your classroom setup, you also can make use of such checkpoints in your teaching in different ways. If your learners have access to digital devices during class, they can use the online version of the book to practice, but you can also use these to generate different versions of a question for group

work or quizzes without devices, or to project a question on the board and automatically verify learners' answers and give feedback.

Syllabus Alignment. The content of this book is aligned with the Kenyan Grade 12 Competency Based Curriculum for Mathematics. In Appendix A and Appendix B you can find a detailed mapping of the content of this book to the curriculum, including which learning outcomes are covered in which sections of the book.

Sections marked with an asterisk (∗) in the title indicate sections that cover content that is not explicitly mentioned in the syllabus, but that we consider important for learners to master.

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Part I

Numbers and Algebra

Chapter 1

Sets

1.1 Set Notation

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Identify set notations and elements of a set in different situations

1.2 Representation of Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Represent sets in different situations

1.3 Types of Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Identify set notations and elements of a set in different situations

1.4 Operations on Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Perform operations on sets in different situations

1.5 Venn Diagrams

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Draw Venn diagrams to represent sets

1.6 Application of Sets

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.1 Sets
Specific Learning Outcomes	Apply sets to real life situations

Chapter 2

Binomial Expansions

2.1 Binomial Expansion

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Binomial Expansions
Specific Learning Outcomes	Expand binomial expressions up to the 4th degree by multiplication

2.2 Coefficients in Binomial Expansions

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Binomial Expansions
Specific Learning Outcomes	Determine coefficients of terms in binomial expansions up to the power of 10

2.3 Evaluating Numerical Cases Using Binomial Expansion

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.2 Binomial Expansions
Specific Learning Outcomes	Evaluate binomial expressions up to the power of 10 in different situations

Chapter 3

Compound Proportions, Rates of Work and Mixtures

3.1 Proportional Parts

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Divide quantities using ratios and proportions in different situations

3.2 Compound Proportions

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Determine direct, indirect and inverse proportions from real life situations

3.3 Rates of Work

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Determine compound proportions and rates of work from real life situations

3.4 Mixtures

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.3 Compound Proportions, Rates of Work and Mixtures
Specific Learning Outcomes	Determine quantities in mixtures in different situations

Chapter 4

Linear Programming

4.1 Forming Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Form linear inequalities from real life scenarios

4.2 Graphing Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Represent linear inequalities on a cartesian plane

4.3 Optimisation of Linear Inequalities

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Determine optimal solutions from inequality graphs

4.4 Application of Linear Programming

Curriculum Alignment

Strand	1.0 Numbers and Algebra
Sub-Strand	1.4 Linear Programming
Specific Learning Outcomes	Apply linear programming to real life situations

Part II

Measurements and Geometry

Chapter 5

Matrices and Transformations

5.1 Transformation Using Matrices

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Draw both the object and its image under a transformation matrix on a Cartesian plane; Determine the matrix of transformation across different situations

5.2 Successive Transformations

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Perform successive transformations of an object using matrices on a Cartesian plane; Determine a single matrix of transformation for successive transformations

5.3 Inverse Transformations

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Determine an inverse of a transformation in different situations

5.4 Area Scale Factor and Determinant of a Matrix

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Relate area scale factor to determinant of a matrix in different situations

5.5 Stretch

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Perform transformation involving shear and stretch in different situations

5.6 Shear

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Perform transformation involving shear and stretch in different situations

5.7 Application of Matrices and Transformations

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.1 Matrices and Transformations
Specific Learning Outcomes	Apply matrix of transformation in different situations

Chapter 6

Area Approximations

6.1 Counting Squares

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Area Approximations
Specific Learning Outcomes	Estimate area of irregular shapes by counting squares technique

6.2 Trapezoidal Rule

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Area Approximations
Specific Learning Outcomes	Determine area bounded by a curve and lines using the trapezoidal rule

6.3 Mid-ordinate Rule

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.2 Area Approximations
Specific Learning Outcomes	Determine area bounded by a curve and lines using the mid-ordinate rule

Chapter 7

Loci

7.1 Perpendicular Bisector Loci

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Relate terms used in loci to normal geometric constructions terms

7.2 Loci of a Point at a Given Distance from a Fixed Point

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Construct loci of points at a given distance from a fixed point

7.3 Loci of a Point at a Given Distance from a Fixed Line

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Construct loci of points at a given distance from a fixed line

7.4 Angle Bisector Loci

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Construct loci of perpendicular line and angle bisectors

7.5 Constant Angle Loci

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Construct loci of points given constant angle in different situations

7.6 Intersecting Loci

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Determine regions enclosed by intersecting loci in different situations

7.7 Loci Involving Inequalities

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	Construct loci involving inequalities in different situations

7.8 Loci Involving Chords

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.3 Loci
Specific Learning Outcomes	[Learning Outcome] (TODO: not yet provided in the curriculum design)

8.1

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Three Dimensional Geometry
Specific Learning Outcomes	Identify skew lines and projection of a line onto a plane

8.2

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Three Dimensional Geometry
Specific Learning Outcomes	Determine the angle between skew lines in an object

8.3

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Three Dimensional Geometry
Specific Learning Outcomes	Calculate the length of a line and projection of line on a plane in a 3-D object

8.4

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Three Dimensional Geometry
Specific Learning Outcomes	Determine the angle between two lines in a 3-D object

8.5

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Three Dimensional Geometry
Specific Learning Outcomes	Determine the angle between a line and a plane in a 3-D object

8.6

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.4 Three Dimensional Geometry
Specific Learning Outcomes	Determine the angle between a plane and a plane in a 3-D object

9.1

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.5 Trigonometry III
Specific Learning Outcomes	Derive the trigonometric identity $\sin^2x+\cos^2x = 1$ from trigonometric ratios

9.2

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.5 Trigonometry III
Specific Learning Outcomes	Draw graphs of trigonometric functions in degrees; Determine amplitude, period, wavelength and phase angle from trigonometric graphs drawn in degrees

9.3

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.5 Trigonometry III
Specific Learning Outcomes	Solve trigonometric equations using the identity $\sin^2x+\cos^2x = 1$ analytically

9.4

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.5 Trigonometry III
Specific Learning Outcomes	Transform trigonometric waves using stretch and translation

10.1

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Express column and position vectors in terms of unit vectors in different situations

10.2

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Express column and position vectors in terms of unit vectors in different situations

10.3

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Express column and position vectors in terms of unit vectors in different situations

10.4

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Determine the magnitude of a vector in three dimensions

10.5

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Determine parallelism of vectors in different situations

10.6

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Perform collinearity test for vectors in two and three dimensions

10.7

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Divide a line proportionally using vectors in different situations

10.8

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Derive and use ratio theorem in vector operations

10.9

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.6 Vectors II
Specific Learning Outcomes	Apply vectors to geometry in different situations

11.1

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.7 Navigation
Specific Learning Outcomes	Identify longitudes and latitude in different situations; Locate points on the earth surface using latitudes and longitudes

11.2

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.7 Navigation
Specific Learning Outcomes	Determine distance along parallels of longitude and latitude in nautical miles and in kilometres

11.3

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.7 Navigation
Specific Learning Outcomes	Determine speed in knots and in kilometres per hour

11.4

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.7 Navigation
Specific Learning Outcomes	Relate longitudes to time at different points on the earth's surface

11.5

Curriculum Alignment

Strand	2.0 Measurements and Geometry
Sub-Strand	2.7 Navigation
Specific Learning Outcomes	Determine the course, track and drift of a body moving in a flowing fluid

12.1

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics II
Specific Learning Outcomes	Determine mean, variance and standard deviation using assumed mean for ungrouped data

12.2

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics II
Specific Learning Outcomes	Draw a cumulative frequency curve for grouped data

12.3

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics II
Specific Learning Outcomes	Calculate median, quartiles, deciles and percentiles for grouped data; Estimate median, quartiles, deciles and percentiles from a cumulative frequency curve

12.4

Curriculum Alignment

Strand	3.0 Statistics and Probability
Sub-Strand	3.1 Statistics II
Specific Learning Outcomes	Determine mean, variance and standard deviation using assumed mean for grouped data; Determine interquartile range and quartile deviation for grouped and ungrouped data analytically and graphically

13.1

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.1 Differentiation II
Specific Learning Outcomes	Determine stationary points and their nature from equations of curves

13.2

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.1 Differentiation II
Specific Learning Outcomes	Sketch curves using stationary points and intercepts

13.3

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.1 Differentiation II
Specific Learning Outcomes	Apply differentiation to maxima and minima in different situations; Apply differentiation in kinematics

14.1

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.2 Integration
Specific Learning Outcomes	Relate integration to differentiation in different situations

14.2

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.2 Integration
Specific Learning Outcomes	Determine indefinite and definite integrals of polynomials up to 3rd degree

14.3

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.2 Integration
Specific Learning Outcomes	Relate integration to the area under a curve in different situations; Determine area under a curve by integration

14.4

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.2 Integration
Specific Learning Outcomes	Relate integration to the area under a curve in different situations

14.5

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.2 Integration
Specific Learning Outcomes	Determine errors from using integration and area approximation methods

14.6

Curriculum Alignment

Strand	4.0 Calculus
Sub-Strand	4.2 Integration
Specific Learning Outcomes	Apply integration to kinematics in different situations

Colophon

This book was authored in Pre-
TeXt by the following contributing team: